Problem Statement-1 :

After JEE Mains, some students got admission into an engineering college. Now there is a class consisting of such n students, and the HOD came to say it is time to select the class monitor. But He never gets all of them at one time. So he brought a register, every time he gets someone with less rank than the previous time he cut the name and wrote the name of the student and the rank.
For a given number of ranks he gets each time, you have to predict how many names are cut in the list.

Constraints:
Number of Visiting<=10^9
ranks <=10000

Input Format:
Number of Visiting N in their first line
N space separated ranks the HOD gets each time

Output Format:
Number of ranks cut in the list

Sample Input:
6
4 3 7 2 6 1

Sample Output:
3

Problem Statement-2 :

Every decimal number can be changed into its binary form. Suppose your computer has it's own CoronaVirus, that eats binary digits from the right side of a number. Suppose a virus has 6 spikes, it will eat up 6 LSB binary digits in your numbers. You will have a bunch of numbers, and your machine will have a virus with n spikes, you have to calculate what will be the final situation of the final numbers.

Input Format:
First line, a single Integer N
Second line N space separated integers of the bunch of values as array V
Third line a single integer n, the number of spikes in Corona for Computer

Output Format:
Single N space separated integers denoting the final situation with the array v.

Sample Input:
5
1 2 3 4 5
2

Output:
0 0 0 1 1

Explanation:
5 is 101 in binary, when you cut the last two binary digits, its 1.

Problem Statement-3 :

Ratul made a linked list, a list made of n nodes, where every node has two variables, the velocity and the mass of a particle.
Since all the particles have the velocity in the same direction, find the total momentum of the entity made by the particles from the linked list.

Constraints :
1<=n<=10000
1<=m,v<=100
Input format:
First line containing n, number of nodes
Then n lines containing the mass and the velocity

space separated.

Output Format:

Single integer denoting the momentum

Sample Input:

4
1 3
2 4
2 3
4 5

Sample Output:

37

Problem Statement –4

A chocolate factory is packing chocolates into the packets. The chocolate packets here represent an array of N number of integer values. The task is to find the empty packets(0) of chocolate and push it to the end of the conveyor belt(array).

Example 1 :

N=8 and arr = [4,5,0,1,9,0,5,0].

There are 3 empty packets in the given set. These 3 empty packets represented as O should be pushed towards the end of the array

Input :

8 – Value of N

[4,5,0,1,9,0,5,0] – Element of arr[O] to arr[N-1],While input each element is separated by newline

Output:

4 5 1 9 5 0 0 0

Example 2:

Input:

6 — Value of N.

[6,0,1,8,0,2] – Element of arr[0] to arr[N-1], While input each element is separated by newline

Output:

6 1 8 2 0 0

Problem Statement –5

You are given a function,
int findCount(int arr[], int length, int num, int diff);

The function accepts an integer array ‘arr’, its length and two integer variables ‘num’ and ‘diff’. Implement this function to find and return the number of elements of ‘arr’ having an absolute difference of less than or equal to ‘diff’ with ‘num’. Note: In case there is no element in ‘arr’ whose absolute difference with ‘num’ is less than or equal to ‘diff’, return -1.

Example:
Input:

- arr: 12 3 14 56 77 13
- num: 13
- diff: 2

Output:
3

Explanation:
Elements of ‘arr’ having absolute difference of less than or equal to ‘diff’ i.e. 2 with ‘num’ i.e. 13 are 12, 13 and 14.

Problem Statement –6

Airport security officials have confiscated several item of the passengers at the security check point. All the items have been dumped into a huge box (array). Each item possesses a certain amount of risk[0,1,2]. Here, the risk severity of the items represent an array[] of N number of integer values. The task here is to sort the items based on their levels of risk in the array. The risk values range from 0 to 2.

Example :

Input :

7 -> Value of N

[1,0,2,0,1,0,2]-> Element of arr[0] to arr[N-1], while input each element is separated by new line.

Output :

0 0 0 1 1 2 2 -> Element after sorting based on risk severity

Example 2:

input : 10 -> Value of N

[2,1,0,2,1,0,0,1,2,0] -> Element of arr[0] to arr[N-1], while input each element is separated by a new line.

Output :

0 0 0 0 1 1 1 2 2 2 ->Elements after sorting based on risk severity.

Explanation:

In the above example, the input is an array of size N consisting of only 0's, 1's and 2s. The output is a sorted array from 0 to 2 based on risk severity.

Problem Statement –7

Mayur is developing a game that has three levels in it. He wants to display the results after the successful completion of all the levels for this purpose he is using a function int giveResult(int s1, s2, s3), where s1, s2, and s3 are the score of every level. This function will return **"Good"** if the s1<s2, **"Best"** if s1>= s2 and s1<=s3 and **"Can do better"** if s1>s3. Can you help Mayur in developing that function

- **Input:**
  - s1 : 10
  - s2 : 12

    - s3 : 15

- **Output :**
    - Good

- **Explanation :**
    - Since 10 is less than 12, hence **“Good”** is returned.

Problem Statement-8:

There are 3 Horses with different speeds(in m/s) in a horse race. The length of the Racing track is given by m meters. You have to calculate the minimum time (in secs.) the fastest horse will win the race.

Input

- The first line of input consists of an integer – *Distance*, representing the length of the racing track.
- The second line of the input consists of 3 space-separated integers representing the speeds of horses running in the race.

Output

- Print the minimum time required to complete the race.

Note:

- Consider all arithmetic operations as integer operations
- Computed values lie within the integer range.

Example

- **Input:**
  - distance: 72
  - Speed of horse1: 8
  - Speed of horse2: 4
  - Speed of horse3: 6

- **Output:**
  - 9

Explanation :

Time is taken by three horses to finish the race are :

- Time of first horse = 72/8 = 9
- Time of second horse = 72/4 = 18
- Time of third horse = 72/6 = 12

The minimum time is 9 secs. Thus, the output is 9

Sample Input

- distance: 18
  - Speed of horse1 : 3
  - Speed of horse2: 9
  - Speed of horse3: 6

- **Sample Output**
  - 2

Problem Statement-9

A Monitor of the class wanted to take the attendance, so he circulated a blank page in the classroom asking students to write there student_Id. But some naughty students wrote their id multiple times to trouble the monitor. Monitor went to the

teacher and reported the incident, the Teacher asked him to mark the absence of students whose id is repeated in the attendance sheet and give him the number of students with repeated id in the attendance sheet.

Input

- The input consists of a string representing the ids of students.

Output

- Print an integer representing the count of characters that are not repeated in the string.

Note: Student id is nothing but an alphabet from a-z.

Example:

- **Input:**
  - ghygbvghyghnhjuhjumnj
- **Output:**
  - 3
- **Explanation:**
  - As there are only 3 students whose ids are not repeating, i.e, d f g v c x and the students with ids repeating are a and s

Problem Statement-10

You have to select an integer number “check” from the range 1 to max and then you have to print the following output after checking that number.

- If the check is a multiple of both 2 and 7, print **“PrepInsta”**
- If the check is a multiple of 2(but not 7), print **“Prep”**
- If the check is a multiple of 7(but not 2), print **“Insta”**
- If i is not a multiple of 2 and 7, print **“Prepster”**

Problem Statement-11

Alex works at a clothing store. There is a large pile of socks that must be paired by color for sale. Given an array of integers representing the color of each sock, determine how many pairs of socks with matching colors there are.

- For example, there are **n=7** socks with colors **ar = {1,2,1,2,1,3,2}**. There is one pair of color **1** and one of color **2**. There are three odd socks left, one of each color. The number of pairs is **2**.

- **Function Description**
  Complete the **sockMerchant** function in the editor below. It must return an integer representing the number of matching pairs of socks that are available.
  sockMerchant has the following parameter(s):
  n: the number of socks in the pile
  ar: the colors of each sock
- **Input Format**
  The first line contains an integer **n**, the number of socks represented in **ar.**
  The second line contains **n** space-separated integers describing the colors **ar[i]** of the socks in the pile.
- **Constraints**
  1 <= n <= 100
  1 <= ar[i] <= 100 & 0 <= i < n
- **Output Format**
  Return the total number of matching pairs of socks that Alex can sell.
- **Sample Input**
  9
  10 20 20 10 10 30 50 10 20

Sample Output

3

- **Explanation**

  Alex can match 3 pairs of socks i.e **10-10, 10-10, 20-20**

  while the left out socks are **50, 60, 20**

Problem Statement-12

Program to check if two given matrices are identical

```
a = [[1,1,1,1],[2,2,2,2],[3,3,3,3],
[4,4,4,4]]
b = [[1,1,1,1],[2,2,2,2],[3,3,3,3],
[4,4,4,4]]
```

Problem Statement-13

You are given a string and your task is to *swap cases*. In other words, convert all lowercase letters to uppercase letters and vice versa.

For Example:

```
Www.HackerRank.com →
wWW.hACKERrANK.COM
Pythonist 2 → pYTHONIST 2
```

Function Description

Complete the *swap_case* function in the editor below.

swap_case has the following parameters:

- *string s:* the string to modify

Returns

- *string:* the modified string

Input Format

A single line containing a string .

Constraints

Sample Input 0

```
HackerRank.com presents "Pythonist 2".
```

Sample Output 0

```
hACKERrANK.COM PRESENTS "pYTHONIST 2".
```

Problem Statement-14

Count Substring in Python

The task is to **count how many times a substring appears in a string**.

Example

Input

```
string: ABCDCDC
substring: CDC
```

Output

```
2
```

Explanation:
`CDC` appears **2 times** in `"ABCDCDC"`.

Problem Statement-15

Mutations (Change a Character in String)

Example

Input

```
abracadabra
5 k
```

Output

```
abrackdabra
```

Problem Statement-16

. Ram is a 6 years old boy who loves to play with numeric legos. One day Ram's mom created a number using those legos and asked ram to tell the number of elements available between two specific numbers 'alpha1' and 'alpha2'. After 15 years when Ram started learning to program, he now wants to write a code to find the number of elements lies between ranges alpha1 and alpha2. If the number is arr and the starting and ending points are alpha1 and alpha2, find the numbers of elements lies in the range.

Input:

- Three space-separated integers

1.
    1. First is the length of arr.
    2. Second is the starting point as alpha1.
    3. The third is the endpoint as alpha2.

- N space-separated array elements.

Output:

- Space-separated indexes between alpha1 and alpha2.

Example:

Input:

9 2 6

1 2 3 4 5 6 7 8 9

Output:

1 2 3 4

Problem Statement-17

. Find the security key to access the bank account in from the encrypted message. The key in the message is the first repeating number from the given message of numbers.

Input:

- String

Output:

- The first repeating number from the message

Example:

Input:

123456654321

Output:

1

Problem Statement-18

In an auditorium, the lighting system of N bulbs is placed in a row. Due to some technical fault in the power supply, only some of the bulbs remain ON and the rest of them go OFF. Due to

this flaw in the network, the bulbs which are OFF are unable to work. So till the technical team finds the actual cause of the fault, the technical head Akshay makes some temporary arrangements for the OFF bulbs at a minimum cost. Akshay decides to connect all the OFF bulbs to the nearby ON bulbs so that the length of the cable used to connect them is minimum. He calculates the distance of the systems from the first system.

Write an algorithm to help Akshay find the minimum length of the cable used to turn all the bulbs ON.

Input

- The first line of the input consists of an integral num, representing the number of bulbs (N).
- The second line consists of N space-separated integers representing the initial state of each bulb, ON(1) or OFF(0)
- The last line consists of N space-separated integers representing the distance of the bulbs from the first bulb.

Output

- Print an integer representing the minimum length of the cable used to turn all the bulbs ON.

Example

- **Input:**
  - 3
  - 1 0 0
  - 1 5 6
- Output:
  - 5

Explanation:

- Length of the cable required to connect the 2nd bulb to the 1st bulb =4
- Length of the cable required to connect the 3rd bulb to the 2nd bulb =1
- The total length of the cable = 5(4+1)

So, the output is 5.